THE FUNDAMENTAL INTUITIONS — EXTENDED

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The Fundamental Intuitions of SAT (Scalar-Angular Torsion Theory)

0th: The "real world" is the world of particle worldlines plotted in four dimensions; it is a hypercomplex tangled "Zottenwelt" of worldlines--SAT calls these worldlines *filaments* (to distinguish from strings in String Theory proper).

1st: Time moves like a wavefront through these worldlines--SAT calls this wavefront *the time surface*.

2nd: What we perceive as a fundamental particle is the intersection of the particle's worldline with the time wavefront.

PILLAR NUMBER ONE: The Standard Model of Particle Physics is the evolving three-dimensional structure of everything--from quarks to organisms to the entire observable universe, and probably beyond. The organizational structure of the universe can be viewed as braided threads, tangled yarns, thick ropes, and ultimately a vast complex and highly structured literal *fabric* of reality, and envisioning these structures and their interactions in four dimensions (or scaled down to three) may give useful intuitive insights into actual physical phenomena.

PROPOSITION A: Unless forced to do otherwise by evidence or the internal logic of SAT, the theory will treat the entirety of the Standard Model of Particle Physics as essentially accurate, and will consider it fully imported into the theory with little or no change, apart from conceptual translation into the geometry of SAT.

3rd: There must be a real interaction between the time wavefront and these worldlines--a transfer of energy from the wavefront to the worldlines, vice versa, or (most likely), both.

4th: These interactions govern particle formation and stability.

5th: These interactions must exert a force back on the wavefront as well.

6th: This back-transmission of energy from the filament network (Zottenwelt) creates curvature of the time surface.

PILLAR NUMBER TWO: The back-pull of the filament structure on the time surface creates curvatures of the time surface that re-create the curved spacetime of General Relativity very naturally from within SAT.

PROPOSITION B: Unless forced to do otherwise by the evidence or the internal logic of the theory, SAT will treat General Relativity as essentially correct, and *fully importable* into SAT's construction of the Standard Model of particle physics with little to no change beyond conceptual translations into the geometry of SAT.

7th: The 4-dimensional extent of a particle's worldline and the time wavefront are physical, not just conceptual or historical objects and therefore, the forces these fundamental structures exert upon one another are quite likely transfered along each filament's extension in the fourth-dimensional extent of the filaments, both forward (in the direction of the wavefront's travel, "futureward"), and backward (in the opposite direction, "pastward"), creating both the possibility the future affecting the present just as the past affects the future. However: See the 8th Fundamental Intuition.

PROPOSITION C: The structure and properties of these fundamental structures (filaments and the time surface) create an intuitive way to understand the limits of these "backbleed" effects; that is, filaments experience tensions, have tensile strength and limitations on their flexibility, and the same should be true of the time surface--thus, backward-propogation in time is structurally self-limiting by the geometry of SAT, the backbleed effects of any tension exerted by interactions in the future-direction are in a sense "baked in" to every interaction as it happens. This would naturally prevent causality violations except perhaps on a very fine--perhaps quantum--scale. This could account for the "wierd" and intuition-breaking implications of quantum physics.

PILLAR NUMBER THREE: Unless forced to do otherwise by the evidence or the internal logic of the theory, SAT will treat the enirety of quantum mechanics as essentially correct, and fully importable into the theory, apart from conceptual translation into the geometry of SAT.

PRIME GENERAL PREDICTION PATHWAY: The tensions exerted upon the past by the future and the future by the past are essentially not accounted for in the Standard Model, General Relativity, or Quantum Mechanics, and this blind spot is likely to produce observable anomolies in the actual behavior of physical systems in the world.

8th: Known observational deviations from the Standard Model, General Relativity, and Quantum

Mechanics may be explainable by general principles derived intuitively from the geometry of SAT.
9th: The nature of the interaction between the time surface and filaments likely encode all of the properties observed in real world particle systems in ways that directly map onto the particle properties and families described by the Standard Model of Particle Physics. As the worldlines of particles (as "hard-coded" into filaments) would likely create complex patterns and structures at the point of intersection with the time surface, it is probable that particle behaviors are described by their intersectional geometry with the wavefront; e.g. the gross or absolute angle of intersection (labeled theta_4) is currently considered to be a primary contributor to imparting mass as an energetic transfer to the filaments from the wavefront--grossly speaking, the greater the angle, the more energy imparted, and the more mass instantiated at the intersection.
PILLAR NUMBER FOUR: The exact geometry of intersection depends also upon the behavior of the filaments; SAT currently treats them as analagous to vibrating guitar strings; the intersectional traces, projected over time of the contact with the wavefront and any given filament should produce helical patterns with variable dimensions and degrees of complexity; this is how SAT interprets the particle interaction laws described by the standard model--for example, quantum chromodynamics interactions are likely to be the result of a bi- and tri-partite limitation on the geometry, possibly modelable by physical coil- or twist-locking mechanisms, spin and flavor are likely mappable to other geometric morphologies present in the SAT geometry. Identifying the structures implicit in this intuition should give new key insights into relationships between the properties that govern the behaviors of particles.
10th: Fundamental physical constants should be natural and intuitive implications about the structure and properties of the SAT geometry or phenomena arising emergently therefrom. For example, the Fine Structure Constant may be an absolute or average tensional limit on the transfer of force from the time surface to the filament network, or a limitation on the torsional limits of the filaments themselves. The Planck Constants should be similarly mappable. Identifying the structural implications of the SAT geometry should provide key insights into the nature and value of these, and other constants.
PROPOSITION D: Light, along with other forces modeled by force-carrying particles (bosons) in

the Standard Model, is likely not truly particulate in nature, but rather phenomena that arise from force-transfer across the filament network by the (fermionic) filaments themselves. SAT currently models light as a sort of "energy transfer release valve" triggered by the absorption of more energy from the time wavefront than the filament's structures and properties can tolerate; a ripple in the filament network.

PROPOSITION E: Because mass is considered to be proportional or approximately proportional to a filament's angle of intersection with the time surface (theta_4), it is proposed that the vacuum of space too is filled with an aligned lattice of filaments whose angle of intersection is near-zero or otherwise very low compared to the filaments that give rise to mass-bearing particles. The low-misalignment allows the time surface to pass with little to no transfer of energy from the time surface to the aligned vacuum filament network. This suggests a number of related interpretations: i. Light travels through the vacuum as localized excitation waves moving roughly perpenducular to the long axes of the aligned filaments. ii. Depending upon the gross large-scale structure of the filament network, there may be multiple vacuum states arising directly from the average regional angle of intersection between this filament lattice and the time surface.

KEY INSIGHT: Both mass-bearing particles and the underlying filamental-latice that fill the vacuum of space are *made of the same fundamental structures* interacting with the time surface in different ways. That is, *both vacuum and matter lie along a single continuum* which defines the local landscape of the intersectional array, and determines the nature of physical systems at any given point in the universe.

PROPOSITION F: Depending upon the exact geometry of intersections and the patterns those intersections trace out when projected over time, it may be that various interpretations of String Theory can be seen as a successful partial-mapping of filament-wavefront geometry. This would allow some version of String Theory to be imported more or less in its entirety with minimal changes apart from translation into the geometric language of SAT.

ADDITIONAL PROPOSITION G: All forces acting upon a particle’s worldline are recorded in that worldline itself; external forces generally impact the worldline vector; internal forces are mapped to superimposed helical oscillations upon the particle worldline.

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EXTENSIONS:

Extension A. The timesheet may be a wavefront of coordinated bosons, a wave of ripples propagating within the filament network.

Extension B. Because SAT uses the same basic mechanism for so many things… That is, the origin of fermions (filament standing waves), bosons (filament vibrations), gravity (large-scale filament distortions) and time (a coordinated wavefront of filament vibrations) all fundamentally being modeled by various forms of distortion in the filament network, they are all, in a significant way, manifestations of the same phenomenon. As such, one should be able to use the same equations for them, just like everyday waves.

Extension C: The trace of vibrating filaments on the time sheet may be equivalent to string theory string identities, which projected over time create complex helical traces in four dimensions; essentially, a string theory vibration mode is what you see when you look down the long axis of a coiled filament.

Extension D: Quantization stems from holonomy.

Extension E: Massless or near-massless particles, like the photon and neutrino, are represented as a single-coil helical excitation of the vacuum filament, and can travel along or across filaments.

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CENTRAL DOGMA

That particle world lines can be mapped into a geometrically defined mathematical structure is trivial. Therefore, to map the entire history of the universe, in principle, is simple. Whether this is treated as a physical reality, or just a conceptual representation of cosmological evolution is not relevant to the fact that this geometrization may provide insights, in that it provides actual *places* in this geometric landscape to look for solutions. The big leap of SAT is simply the choice to treat this already-useful map as having a physical reality of its own, and suggesting that therefore, forces within the “zottenwelt” might be transmitted along and between filaments, as well as along and between the timesheet and filaments.

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Tentative:

Photons might be interpreted as identical to a single-coil neutrino helix, but partially rotated in the fourth dimension and thereby, an extension of neutrino flavor oscillation; as the coil precesses, it may rotate such that its neutrinolike properties become inaccessible to three-space interactions, appearing to drop away.

The oscillation process may explain or partly explain dark matter—neutrino/photon oscillation may continue until the particle appears to ‘blink out’ of our dimensional slice; thereby providing a novel explanation of dark matter: That fraction of the universe whose light has become invisible to us through ‘blinking out’.